aminoacids). Growth rate can be controlled by adjusting the dilution rate and cell density is controlled by modifying the concentration of the limiting nutrient (Figure 6.5).

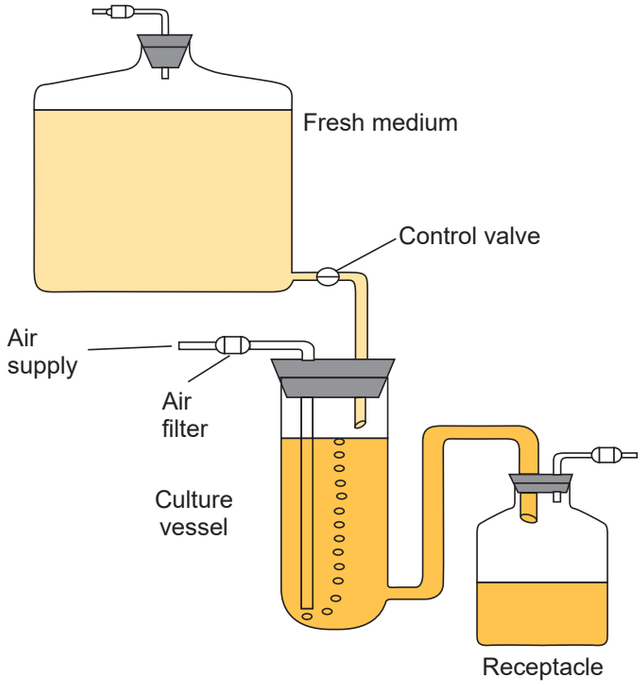


Figure 6.5: The Chemostat

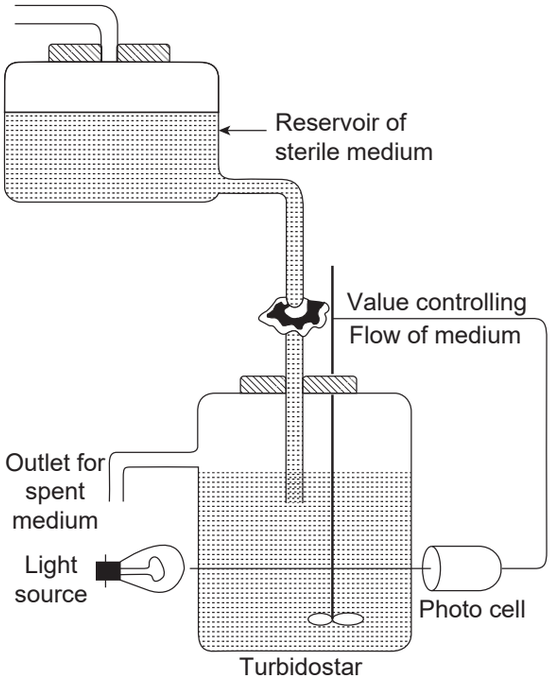


Figure 6.6: The Turbidostat

Turbidostat

This type of continuous culture system has a photocell that measures the turbidity of the culture vessel. This automatically regulates the flow rate of the culture medium. Turbidostat does not contain limiting nutrients (Figure 6.6).

6.5.1 Factors Influencing Growth

The growth and activities of microorganisms are greatly influenced by the physical and chemical conditions of their environment. Among all factors, four key factors play major roles in controlling the growth of microorganisms. They are

1. Temperature
2. pH
3. Water activity
4. Oxygen

1. Temperature

Temperature is one of the most important environmental factor affecting the growth and survival of microorganisms. Temperature can affect microorganisms because the enzyme catalysed reactions are sensitive to fluctuations in temperature.

For every microorganism, there is a minimum temperature below which no growth occurs, an optimum temperature at which growth is most rapid, and a maximum temperature above which no growth occur. These three temperatures are called cardinal temperatures.

Temperature classes of microorganisms

Microorganisms are broadly distinguished into four groups in relation to their temperature optima.

- Psychrophiles

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